

GR00T N1 · arXiv 2503.14734 · Oct 7 club

Empiricist

11+3 · seat empty

H1: K=4 near-optimal · K/H smoke grid

Tattoo numbers · ruthless Defer

Spine: Eagle-2 System 2 (~10 Hz) + FM DiT System 1 (120 Hz) via cross-attn

H=16 · K=4 · N1 $\neq$ N1.7 · NOT a classical planner

NVIDIA · Fourier GR-1 · open humanoid VLA

Tattoo numbers · remember these

Table 2 sim @100 demos · Table 3 real GR-1 · preserve DexMG mismatch

SIM avg @100 (Table 2)

N1-2B ~45%

vs DP 33.4% · BC-T 26.4%

RoboCasa 32.1 · DexMG 66.5 · GR-1 50.0

App Table 4 DexMG@100 = 58.5 $\neq$ 66.5

REAL GR-1 avg (Table 3)

N1 full 76.8%

vs DP full 46.4%

N1 @10% $\rightarrow$ 42.6% ($\approx$ DP full -3.8 pp)

$n \approx 10$ trials · Pack Machinery $n=5$

Systems

63.9 ms · H=16 · K=4 · L40

Pretrain

$\sim 50k$ H100-h · BS 16384

Neural

88h $\rightarrow$ 827h I2V · Defer factory

Empiricist: never quote Partial one-task smoke as Table 2 average

Hypotheses · feasible track

H1 NFE

K=4 near-optimal; K=1 hurts; flatten by ~4-8

H2 data

Adapter FT from N1 beats scratch DP @ 30 demos

H3 mid-layer

Layer-12 $\geq$ final for SR and/or latency

H4 forget

Narrow FT erases pretrained bimanual strategies

H5 π cousin

Give CFM math; need adapter wiring $\neq$ OpenPI drop-in

H6 RL locus

Polish ID skills — won't invent long-horizon

Empiricist smoke grid · eval-only NFE × horizon

H1: K=4 near-optimal · flatten by ~4-8 · Partial ≠ Table 2 averages

K \ H	H=8	H=16 ★N1	H=32	H=40 N1.7
K=1	hurt SR?	K=1 fail	—	—
K=2	smoke	P0	Partial	N1.7?
K=4	Partial	★ paper OP	Partial	N1.7 try
K=8	lat ↑	flat?	lat ↑	lat ↑

P0 Y

ckpt dry infer · $K \in \{1, 2, 4, 8\}$ eval grid

Defer N

full pretrain ~50k H100-h · full RoboCasa×GR-1 · real GR-1

Priority matrix · ruthless smoke rule

P0 Y

Released N1-2B dry infer; eval-only NFE/chunk grid

P0 Partial

Mid-layer vs final VL features (needs hook)

P1 Partial

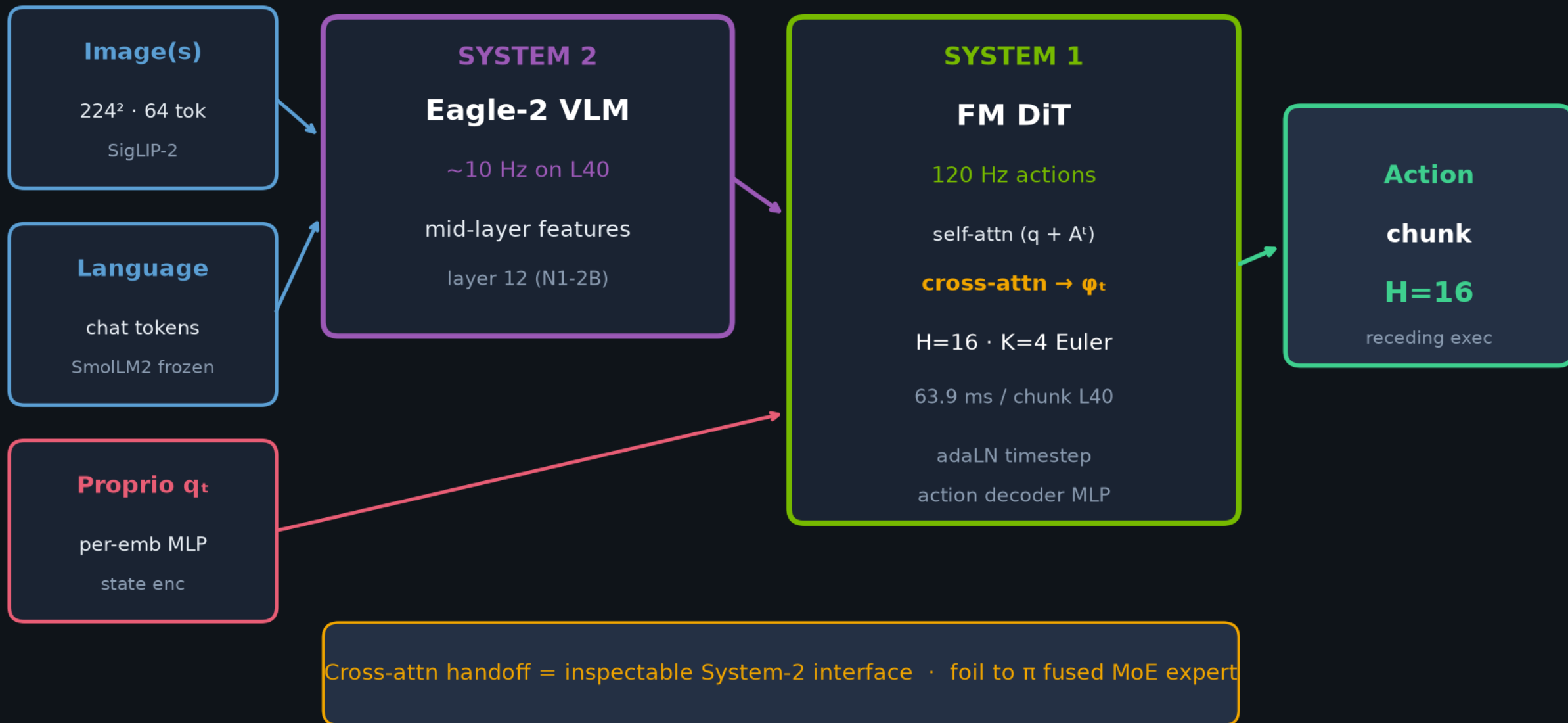
Adapter-only FT 30–100 demos; data-regime curve

Defer N

Full pretrain · WAN factory · full matrix · real GR-1 · N1.7 from scratch

Dual-system VLA · Eagle-2 System 2 × FM DiT System 1

Jointly trained end-to-end · cross-attn bridge · NOT a classical planner



Control spine · CFM cousin, dual-system packaging

NOT classical planner · generative continuous chunk controller

$$A_t^\tau = \tau A_t + (1 - \tau)\varepsilon \quad \cdot \quad \mathcal{L}_{fm} = \mathbb{E} \|V_\theta - (\varepsilon - A_t)\|^2$$

VL @ 10 Hz

Eagle-2 mid features
cache across denoise

Denoise K=4

Euler on DiT
63.9 ms / H=16

Commit + replan

Don't wait full H
open-loop; recede

Contrast: OpenPI action-expert FM · DP CNN/Transformer · π_{RL} PPO · Comet RFT

Same CFM math as π_0 — different reasoning↔motor handoff

VERSION DISCIPLINE · N1 ≠ N1.7

APIs / horizons / backbones differ — pin the checkpoint you serve

Paper N1

GR00T-N1-2B · ~2.2B (1.34B VLM)

Eagle-2 · SigLIP-2 + SmolLM2

H=16 · K=4 · mid-layer 12

63.9 ms / chunk @ L40 bf16

Table 2/3 tattoo numbers live [HERE](#)

N1.7 GA

nvidia/GR00T-N1.7-3B · ~3B

Cosmos-Reason2-2B (Qwen3-VL)

H ≤ 40 · new package / API

Apache-2.0 · BEHAVIOR'26 baseline

Do NOT quote N1 tables as N1.7

Stakeholder / Visionary rule

Prefer finetune N1.7 for product/challenge · use N1-2B only for paper replication

Match served horizon to trained horizon · N1.6 BEHAVIOR1k ≠ N1.7 forecast

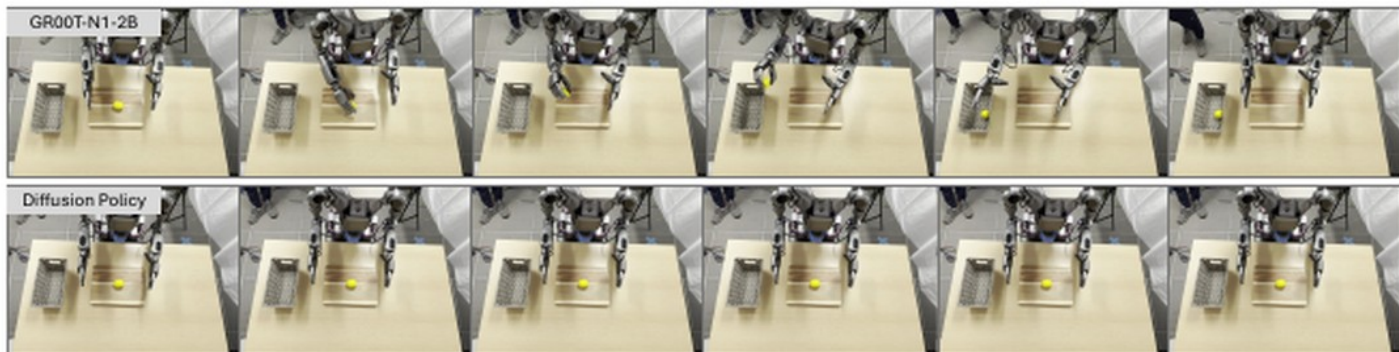
Also: N1 is NOT a classical planner — generative CFM chunk controller with dual-system packaging

Task: Pick up red apple and place it into the basket

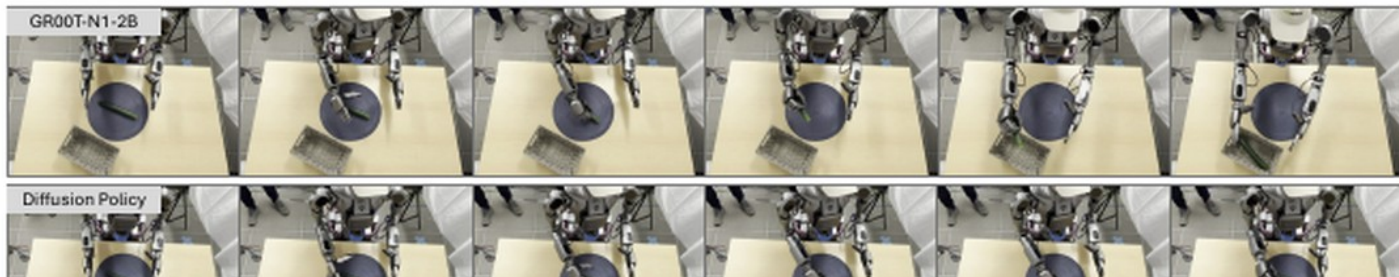


Figure 11: Pre-training Qualitative Example. While prompting the pretrained GR00T-N1-2B model with a post-training task instruction, we even increase the difficulty by placing the apple to the left of both hands. Despite not having encountered this setup during training, the model successfully places the red apple into the basket via a two-handed handover, albeit with jerkier motion.

Task: Pick up lemon from cutting board to the basket



Task: Pick up cucumber from placemat to the basket

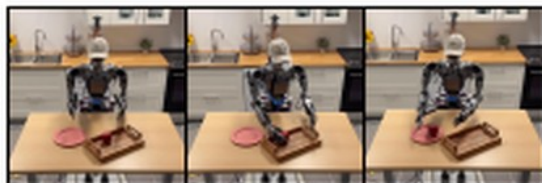


bottom shelf

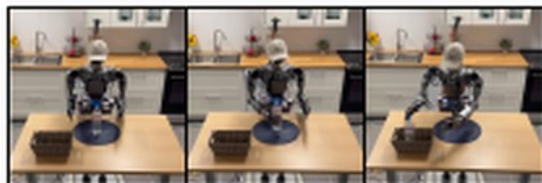
Pick-and-Place with Left-to-Right Handover

Pick up Novel Object to Novel Container

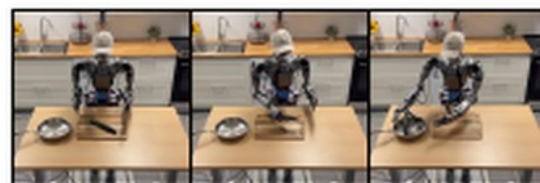
Post-Training Evaluations



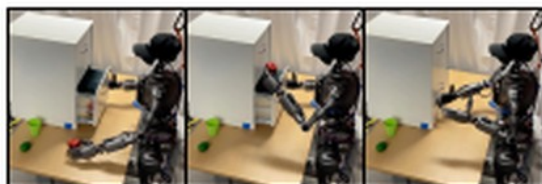
Pick-and-Place: Tray to Plate



Pick-and-Place: Placemat to Basket



Pick-and-Place: Cutting Board to Pan



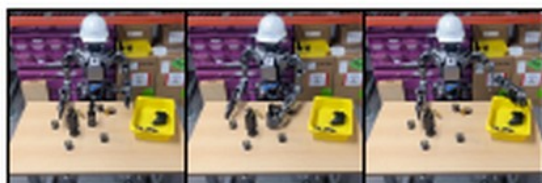
Articulated: White Drawer



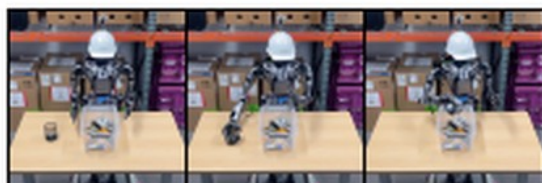
Articulated: Wooden Chest



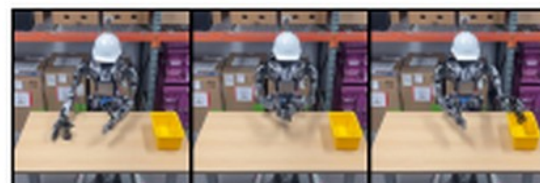
Articulated: Dark Cabinet



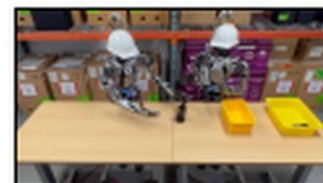
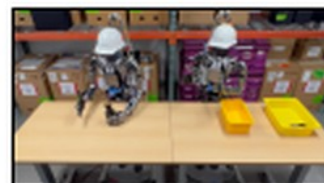
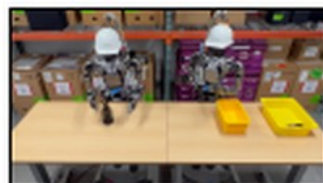
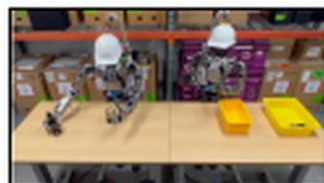
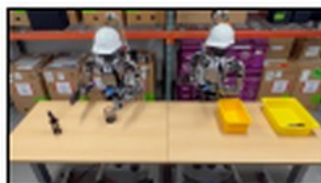
Industrial: Machinery Packing



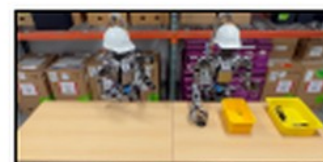
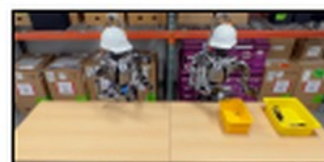
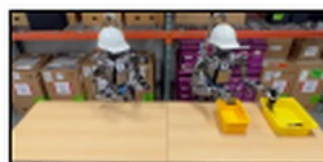
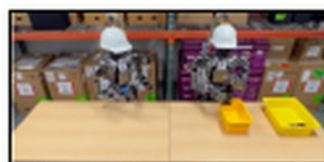
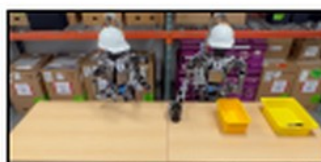
Industrial: Mesh Cup Pouring



Industrial : Cylinder Handover



Coordination Part 1: Cylinder to Mesh Cup and Mesh Cup Handover to Another Robot



GR00T N1 vs π_0 / $\pi_{0.5}$ · same CFM family, different handoff

BEHAVIOR'26 ships BOTH as official baselines — dual-track is first-class

AXIS	GR00T N1	π_0 / $\pi_{0.5}$
Coupling	Separate DiT · cross-attn	Action expert · shared self-attn
VLM	Eagle-2 (SmolLM2+SigLIP-2)	PaliGemma lineage
Chunk / denoise	H=16 · K=4 · 10/120 Hz	H=50 (π_0) · ~10 Euler · 20-50 Hz
Embodiment I/O	Per-emb MLPs · LAPA/IDM	Pad/mask shared dim
Data bet	Pyramid + neural I2V	OXE-scale + $\pi_{0.5}$ co-train
Successor	N1.7 ~3B · H≤40 · Apache-2.0	$\pi_{0.5}$ + RTC + BEHAVIOR forks

FLAG: modular dual-system / cross-attn DiT vs fused shared-attn MoE expert · same chunk+flow interface

Empiricist close

Smoke first · Defer the 50k H100-h pretrain

H1: K=4 · report NFE/H · Partial $\neq$ Table 2

GR00T N1 · Oct 7 · all seats empty · still full five packs

Data pyramid · quantity ↓ · embodiment-specificity ↑

~593M frames / ~8376 h · human + neural + DexMimicGen + real

PEAK · Real robot

GR-1 teleop ~88 h · OXE · AgiBot-Alpha · RH20T

MIDDLE · Synthetic

DexMimicGen ~780k traj · neural I2V 88h→827h · LAPA/IDM

BASE · Web + human egocentric

Ego4D · Ego-Exo4D · EPIC · HOI4D · Assembly-101 · LAPA latent actions

Visionary durable idea: open humanoid data pyramid survives backbone swaps (Eagle → Cosmos-Reason2 → ...)

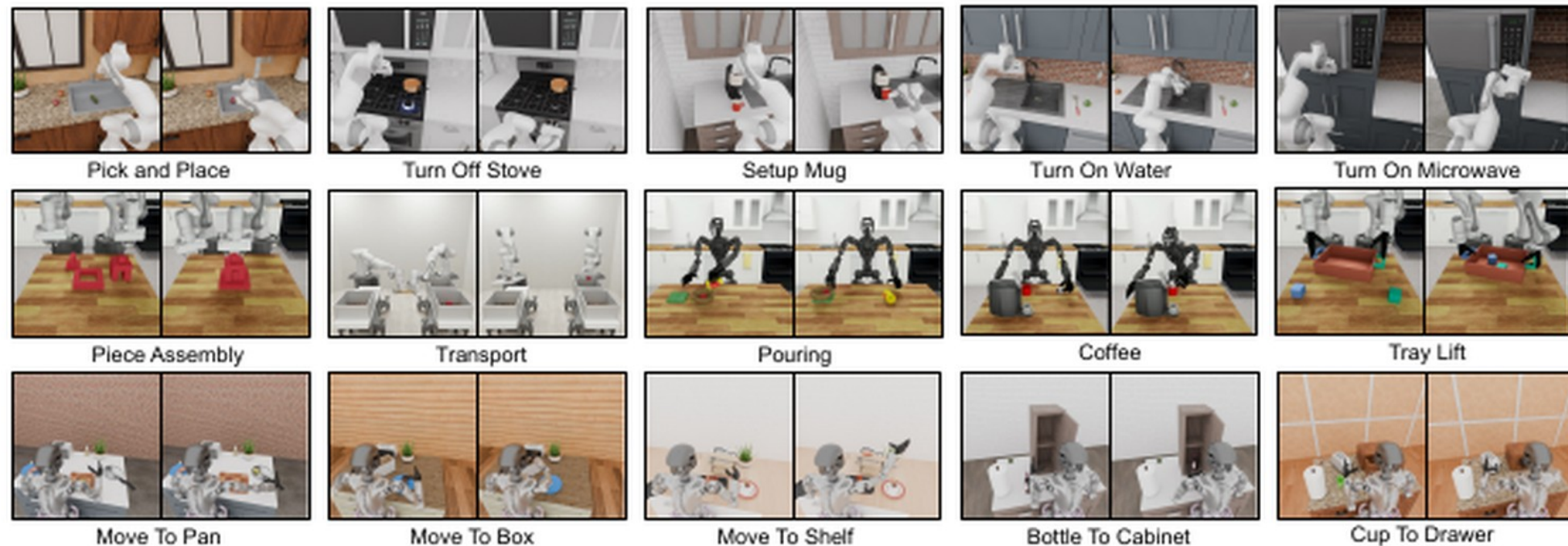


Figure 7: **Simulation Tasks.** Our simulation experiments use tasks from two open-source benchmarks (RoboCasa (Nasiriany et al., 2024) in the top row and DexMimicGen (Jiang et al., 2024) in the middle row) and a newly developed suite of tabletop manipulation tasks that closely resemble our real-world tasks (bottom row). We provide Omniverse renderings of the tasks above.

first-person perspectives of hand-object interactions (examples shown in Figure 14). Our video datasets include the following:

- **Ego4D** is a large-scale egocentric video dataset that includes diverse recordings of everyday activi-